

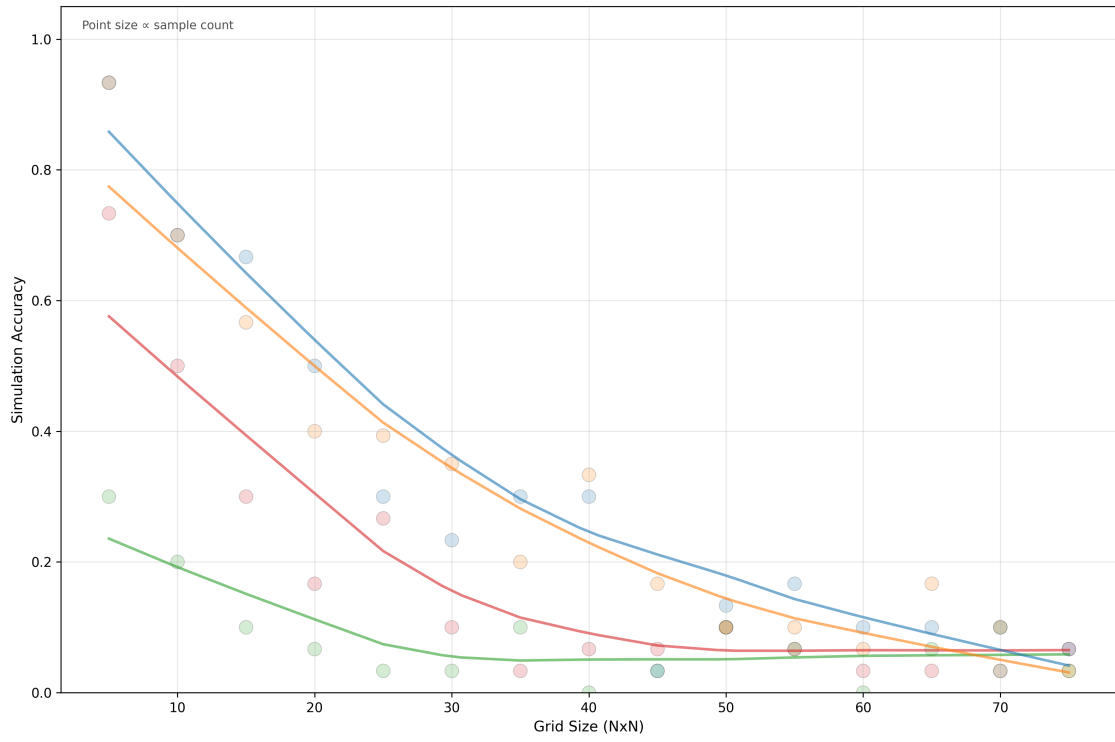


Figure 16: Accuracy vs Grid Size (Slide)

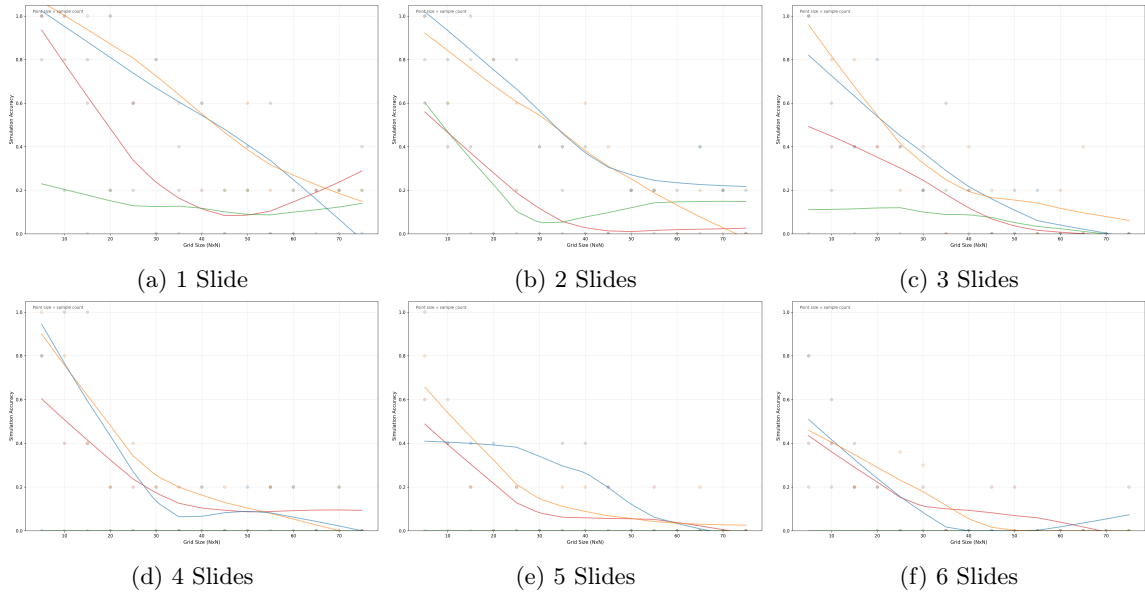


Figure 17: Accuracy vs Grid Size per Slide Count (Slide)

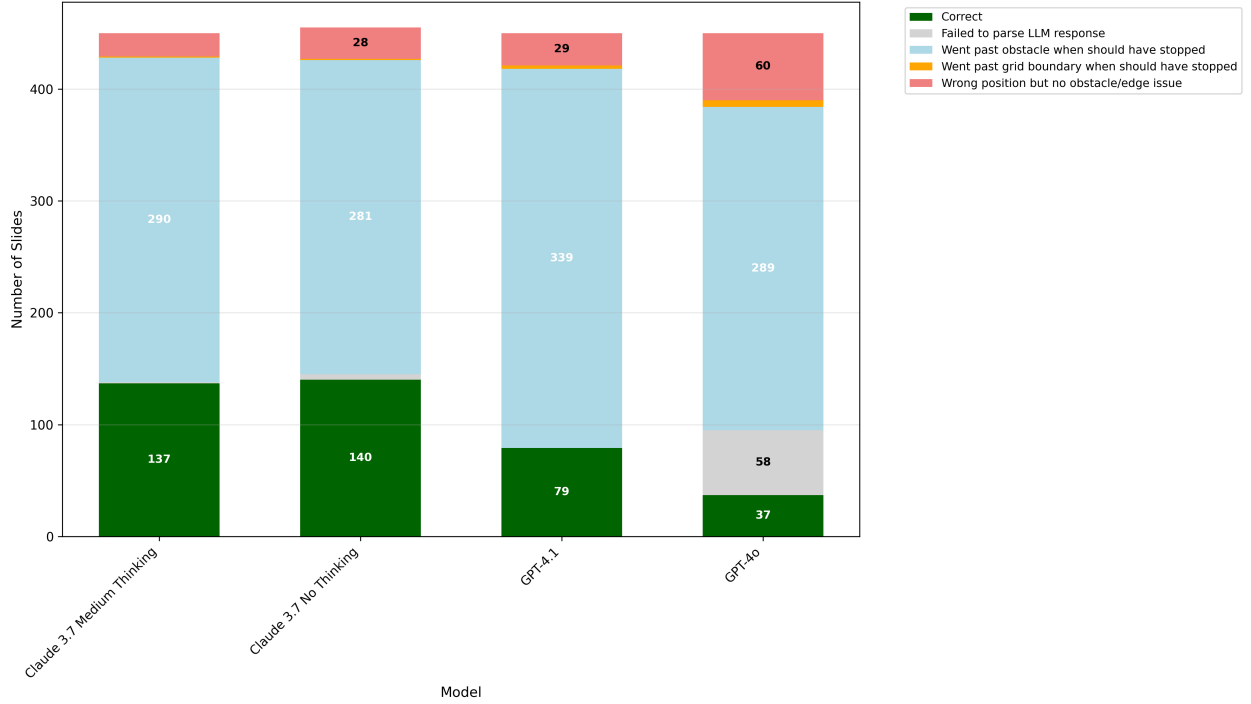


Figure 18: Error Types (Slide)

4.6 Parsing

We also encountered some parsing errors with the LLM responses despite efforts with both prompting and regex to make the parsing as reliable as possible. GPT-4o showed significantly higher rates of parsing errors, with 10% of the total responses in an unparseable or incomplete format. Parsing error rates also increased with grid size, demonstrating once again the further degeneration of model response reliability as complexity increases. These phenomena are seen in Figures 19 and 20.

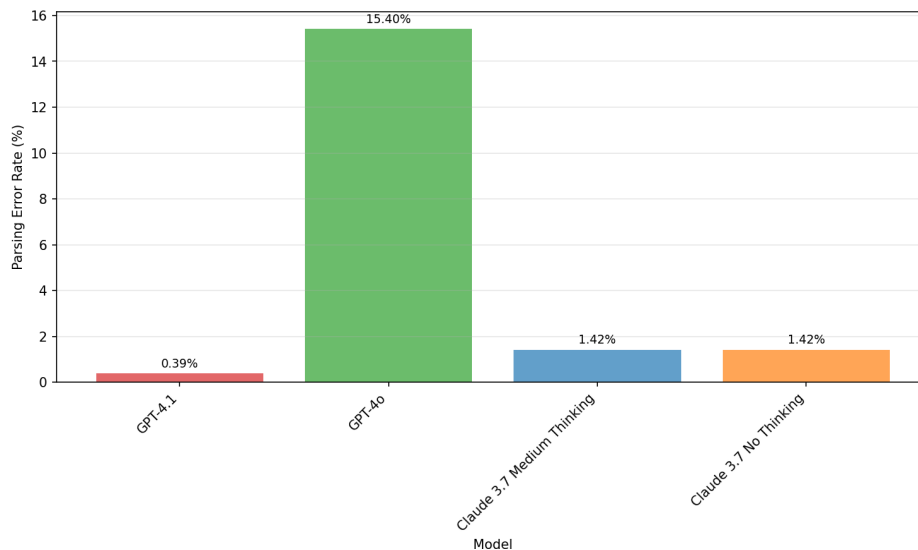


Figure 19: Parsing Error Rates (All tests)

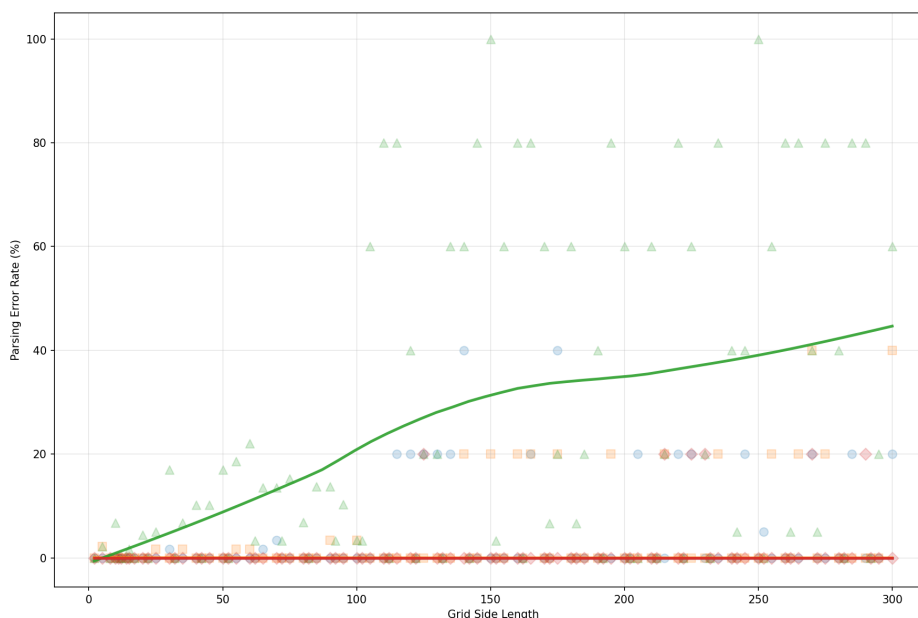


Figure 20: Parsing Error vs Grid Size (All tests)

4.7 Tokenization Changes

In an attempt to mitigate the loss of accuracy, we experimented with several layouts and tokenizations of the grid. Changes were tested on the Quadrant and Search tests. Quadrant was selected because as it was the most internally consistent of the tests, meaning that most problems lay with interpretation of the initial board. Search, on the other hand, was the most difficult, and therefore had the most room for improvement. New grid formats can be checked for tokenization with GPT-4o here, with GPT-4.1 using the same tokenizer. The tokenizer for the Anthropic models is private and therefore unavailable. As such, it should be noted that